In [10]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error,r2_score
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import PolynomialFeatures
```

In [4]:
```
df = pd.read_csv('/home/backup/ml_datasets/auto_mpg.csv');
print(df.head(20))
```

```
     mpg  cylinders  displacement  horsepower  weight  acceleration  \
0   18.0          8         307.0       130.0    3504          12.0
1   15.0          8         350.0       165.0    3693          11.5
2   18.0          8         318.0       150.0    3436          11.0
3   16.0          8         304.0       150.0    3433          12.0
4   17.0          8         302.0       140.0    3449          10.5
5   15.0          8         429.0       198.0    4341          10.0
6   14.0          8         454.0       220.0    4354           9.0
7   14.0          8         440.0       215.0    4312           8.5
8   14.0          8         455.0       225.0    4425          10.0
9   15.0          8         390.0       190.0    3850           8.5
10  15.0          8         383.0       170.0    3563          10.0
11  14.0          8         340.0       160.0    3609           8.0
12  15.0          8         400.0       150.0    3761           9.5
13  14.0          8         455.0       225.0    3086          10.0
14  24.0          4         113.0        95.0    2372          15.0
15  22.0          6         198.0        95.0    2833          15.5
16  18.0          6         199.0        97.0    2774          15.5
17  21.0          6         200.0        85.0    2587          16.0
18  27.0          4          97.0        88.0    2130          14.5
19  26.0          4          97.0        46.0    1835          20.5

    model_year  origin                       name
0           70     usa  chevrolet chevelle malibu
1           70     usa          buick skylark 320
2           70     usa         plymouth satellite
3           70     usa              amc rebel sst
4           70     usa                ford torino
5           70     usa           ford galaxie 500
6           70     usa           chevrolet impala
7           70     usa          plymouth fury iii
8           70     usa           pontiac catalina
9           70     usa         amc ambassador dpl
10          70     usa        dodge challenger se
11          70     usa         plymouth 'cuda 340
12          70     usa      chevrolet monte carlo
13          70     usa    buick estate wagon (sw)
14          70   japan      toyota corona mark ii
15          70     usa            plymouth duster
16          70     usa                 amc hornet
17          70     usa              ford maverick
18          70   japan               datsun pl510
19          70  europe  volkswagen 1131 deluxe sedan
```

In [5]:
```
df = df.drop('name',axis = 1)
```

In [6]:
```
print(df.head(20))
```

```
     mpg  cylinders  displacement  horsepower  weight  acceleration  \
0   18.0          8         307.0       130.0    3504          12.0
```

```
1   15.0          8         350.0       165.0    3693          11.5
2   18.0          8         318.0       150.0    3436          11.0
3   16.0          8         304.0       150.0    3433          12.0
4   17.0          8         302.0       140.0    3449          10.5
5   15.0          8         429.0       198.0    4341          10.0
6   14.0          8         454.0       220.0    4354           9.0
7   14.0          8         440.0       215.0    4312           8.5
8   14.0          8         455.0       225.0    4425          10.0
9   15.0          8         390.0       190.0    3850           8.5
10  15.0          8         383.0       170.0    3563          10.0
11  14.0          8         340.0       160.0    3609           8.0
12  15.0          8         400.0       150.0    3761           9.5
13  14.0          8         455.0       225.0    3086          10.0
14  24.0          4         113.0        95.0    2372          15.0
15  22.0          6         198.0        95.0    2833          15.5
16  18.0          6         199.0        97.0    2774          15.5
17  21.0          6         200.0        85.0    2587          16.0
18  27.0          4          97.0        88.0    2130          14.5
19  26.0          4          97.0        46.0    1835          20.5

    model_year  origin
0           70     usa
1           70     usa
2           70     usa
3           70     usa
4           70     usa
5           70     usa
6           70     usa
7           70     usa
8           70     usa
9           70     usa
10          70     usa
11          70     usa
12          70     usa
13          70     usa
14          70   japan
15          70     usa
16          70     usa
17          70     usa
18          70   japan
19          70  europe
```

In [7]:

```
X=df[['displacement']]
y=df[['mpg']]

X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random_state=42)

print(f"X_train shape: {X_train.shape[0]}")
print(f"X_test shape: {X_test.shape[0]}")
print(f"y_train shape: {y_train.shape[0]}")
print(f"y_test shape: {y_test.shape[0]}")
```

```
X_train shape: 318
X_test shape: 80
y_train shape: 318
y_test shape: 80
```

In [12]:

```
poly=PolynomialFeatures(degree=2)
X_train_poly=poly.fit_transform(X_train)
X_test_poly=poly.transform(X_test)
```

In [22]:

```
poly_reg = LinearRegression()
poly_reg.fit(X_train_poly,y_train)
y_pred = poly_reg.predict(X_test_poly)
```

In [23]:
```
print(f"Intercept: {poly_reg.intercept_}")
print(f"Coefficient: {poly_reg.coef_}")
```

```
Intercept: [42.19890516]
Coefficient: [[ 0.         -0.13827949  0.00016763]]
```

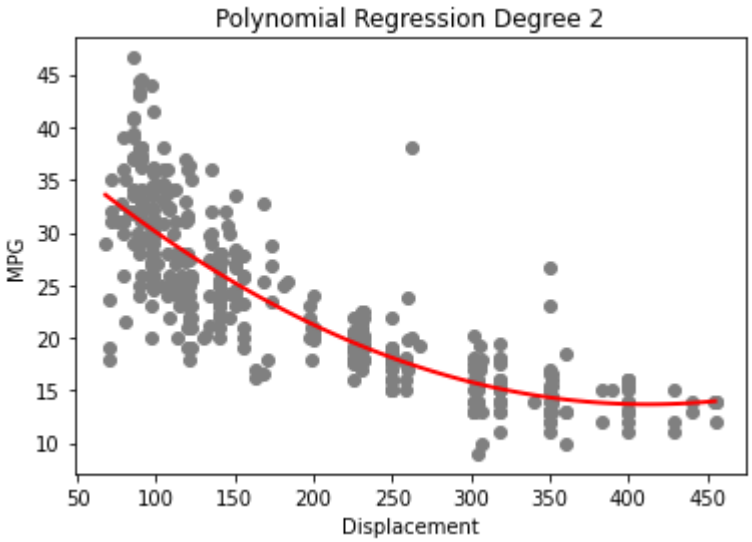
In [24]:
```
mse=mean_squared_error(y_test,y_pred)
print(f"Mean Squared Error: {mse}")

#claculate r squared score
r2=r2_score(y_test,y_pred)
print(f"R-squared score: {r2}")
```

```
Mean Squared Error: 15.10735356108076
R-squared score: 0.7190189176110284
```

In [27]:
```
plt.scatter(X,y,color='grey')
x_range = pd.DataFrame(np.linspace(X['displacement'].min(),X['displacement'].max(),1

x_range_poly=poly.transform(x_range)
y_range=poly_reg.predict(x_range_poly)
plt.plot(x_range,y_range,color="red",linewidth=2,label="DEGREE 2")
plt.xlabel("Displacement")
plt.ylabel("MPG")
plt.title("Polynomial Regression Degree 2")
plt.show()
```



In [29]:
```
poly5 = PolynomialFeatures(degree = 5)
x_train_poly5 = poly5.fit_transform(X_train)
x_test_poly5 = poly5.transform(X_test)

poly_reg5 = LinearRegression()
poly_reg5.fit(x_train_poly5, y_train)
y_pred5 = poly_reg5.predict(x_test_poly5)
```

In [30]:
```
print(f"Intercept: {poly_reg5.intercept_}")
print(f"Coefficient: {poly_reg5.coef_}")
```

```
Intercept: [29.50365643]
Coefficient: [[ 0.00000000e+00  2.34968100e-01 -3.74855062e-03  1.83316313e-05
  -3.89543691e-08  3.05815095e-11]]
```

In [31]:
```
mse=mean_squared_error(y_test,y_pred5)
print(f"Mean Squared Error: {mse}")

#claculate r squared score
r2=r2_score(y_test,y_pred5)
print(f"R-squared score: {r2}")
```

```
Mean Squared Error: 15.230931613467419
R-squared score: 0.7167204942122025
```

In [33]:
```
poly7 = PolynomialFeatures(degree = 7)
x_train_poly7 = poly7.fit_transform(X_train)
x_test_poly7 = poly7.transform(X_test)

poly_reg7 = LinearRegression()
poly_reg7.fit(x_train_poly7, y_train)
y_pred7 = poly_reg7.predict(x_test_poly7)
```

In [34]:
```
print(f"Intercept: {poly_reg7.intercept_}")
print(f"Coefficient: {poly_reg7.coef_}")
```

```
Intercept: [36.19012864]
Coefficient: [[ 0.00000000e+00 -5.18292003e-09 -2.51389230e-07 -1.64373543e-05
   1.52876548e-07 -5.73571293e-10  9.89710929e-13 -6.49797568e-16]]
```

In [35]:
```
mse=mean_squared_error(y_test,y_pred7)
print(f"Mean Squared Error: {mse}")

#claculate r squared score
r2=r2_score(y_test,y_pred7)
print(f"R-squared score: {r2}")
```

```
Mean Squared Error: 14.919468839579093
R-squared score: 0.7225133782522284
```

In [37]:
```
model = LinearRegression()
model.fit(X_train, y_train)
print(f"Intercept: {model.intercept_}")
print(f"Coefficient: {model.coef_}")
```

```
Intercept: [35.45349528]
Coefficient: [[-0.06172521]]
```

In [38]:
```
y_pred_linear=model.predict(X_test)
mse=mean_squared_error(y_test,y_pred_linear)
print(f"Mean Squared Error: {mse}")

#claculate r squared score
r2=r2_score(y_test,y_pred_linear)
print(f"R-squared score: {r2}")
```

```
Mean Squared Error: 18.102543998358946
R-squared score: 0.6633114869465596
```

In [53]:
```
plt.scatter(X,y,color='lightblue')
x_range = pd.DataFrame(np.linspace(X['displacement'].min(),X['displacement'].max(),1

#degree 2
x_range_poly=poly.transform(x_range)
```

```
y_range=poly_reg.predict(x_range_poly)
plt.plot(x_range,y_range,color="orange",linewidth=2,label="DEGREE 2")

#degree 5
x_range_poly5=poly5.transform(x_range)
y_range5=poly_reg5.predict(x_range_poly5)
plt.plot(x_range,y_range5,color="blue",linewidth=2,label="DEGREE 5")

#degree 7
x_range_poly7=poly7.transform(x_range)
y_range7=poly_reg7.predict(x_range_poly7)
plt.plot(x_range,y_range7,color="green",linewidth=2,label="DEGREE 7")


#linear regression
plt.plot(X_test[:50],y_pred_linear[:50],color='yellow', label= "LinearRegression")

plt.xlabel("Displacement")
plt.ylabel("MPG")
plt.title("Polynomial Regression Degree 2")
plt.show()
```

